

Beyond Speed Rankings: Application-Layer Broadband Performance in Nine Southeast Asian Countries

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ABSTRACT

Southeast Asian broadband is usually judged by headline speed-index rankings, which say little about whether real applications work. Thailand ranked 12th worldwide for fixed broadband in June 2026, yet a 2023 national consumer survey found 81% of 2,924 respondents reporting connection problems [7]. We analyze 990 million speed tests — 229.4M from Ookla Open Data and 760.6M from M-Lab NDT7, Q1 2023 to Q4 2025 — across nine countries, and compute the test-weighted share of province-quarters that clear the bandwidth (and, for cloud gaming, latency) requirements of voice, HD video, UHD video and cloud gaming (both datasets are crowdsourced and user-initiated rather than a random sample). Voice and HD video clear their bars in essentially every province-quarter across all twelve quarters. Fixed broadband stratifies sharply by geography, with a 16× spread between the strongest and weakest capital regions. Against the combined cloud-gaming bar, the binding constraint flips by market: low-tier fixed broadband (Cambodia, Laos, Myanmar) fails mostly on bandwidth, while mobile networks in every single country — including Singapore, at only 48.8% — fail mostly on latency. Work is ongoing on per-province modeling and on peak-hour behavior.

KEYWORDS

Internet measurement, broadband performance, Southeast Asia, Ookla, M-Lab NDT7, application requirements

1 MOTIVATION

National broadband quality in Southeast Asia is usually argued from two sources that disagree with each other: index rankings and user complaints. Thailand was ranked 12th worldwide for fixed broadband in Ookla’s Speedtest Global Index in June 2026 [5], while a 2023 survey by the Thailand Consumer Council found that 81% of 2,924 respondents had encountered connection problems, mostly slow or dropping connections [7]. A national mean download speed cannot reconcile those two observations, because it describes the network rather than the service a user is trying to run over it.

We therefore ask a different question: *how do Southeast Asian networks perform against the demands of the applications people actually use?* Rather than ranking countries by speed, we score every country-quarter against published bandwidth requirements and report how close each country’s median connection comes to clearing each bar. The mismatch above is not unique to Thailand, so we run the analysis across nine countries.

2 DATA AND METHOD

We use two complementary open datasets covering Thailand, Vietnam, Laos, Myanmar, Cambodia, Indonesia, the Philippines, Malaysia and Singapore, over twelve quarters (Q1 2023–Q4 2025):

- **Ookla Open Data** [4] — 229,367,221 tests, published as quarterly tiles of roughly 600 m×600 m, fixed and mobile. Ookla servers sit inside access networks and use parallel streams, so these tests characterize last-mile capacity.
- **M-Lab NDT7** [3] — 760,607,387 single-stream tests to off-net servers, closer to what a single streaming or browsing flow observes.

Tiles are assigned to first-level administrative provinces by point-in-polygon against geoBoundaries ADM1 [6]; NDT7 points are joined to the same polygons. Province averages are test-weighted, and we keep only province-quarters with at least 100 tests and at least 5 tiles. Tests with non-positive throughput or duration are dropped, and IP addresses are checked against ip-api.com to exclude hosting/proxy traffic. Because user-initiated speed tests are a biased sample [1], we report distributions and success rates rather than treating any single figure as a population mean. Coverage is uneven: Myanmar is missing 10.7% of fixed and 22% of mobile province-quarters, and some Vietnamese and Cambodian provinces lack mobile data in low-volume quarters — gaps we flag rather than interpolate.

We evaluate four application classes against the thresholds of Lübben and Misfeld [2]: voice (64 kbps, 200 ms), HD video streaming (5 Mbps), UHD video streaming (25 Mbps) and cloud gaming (44 Mbps, 25 ms). **Pass rate** is the test-weighted share of province-quarters (fixed and mobile separately, all twelve quarters, Q1 2023–Q4 2025) whose average download speed — and, where a latency bound applies, average latency — clears the threshold; this is an aggregate over 100+ tests per province-quarter, not a per-test rate. Latency is evaluated using **Ookla**, not M-Lab NDT7: for six of our nine countries most NDT7 tests terminate at an offshore server (Thai tests mainly in Singapore or Hong Kong), so NDT7 round-trip time mostly reflects international transit rather than domestic access, whereas Ookla selects a nearby server, matching our faculty advisor’s ruling that cloud-gaming latency be evaluated on Ookla. Ookla latency is still an idle measurement to an often on-net server, so it is a lower bound on the round-trip time a real interactive session would see.

3 PRELIMINARY RESULTS

3.1 Voice and HD video are effectively solved

The voice (64 kbps) and HD-video (5 Mbps) bandwidth bars pass in 100% of province-quarters, fixed and mobile, in every one of the

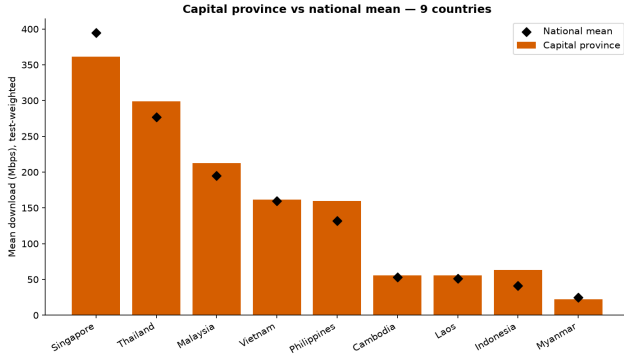


Figure 1: Capital province versus national mean download speed, fixed broadband, test-weighted, nine countries.

nine countries, across all twelve quarters. Voice’s 200 ms latency bar is just as uniform, with one exception: Myanmar’s fixed-broadband province-quarters pass it only 98.2% of the time (mobile: 99.1%). Whatever the regional digital divide consists of, it is not a failure to carry a phone call or a 5 Mbps video stream — though this says nothing about the tail of individual tests.

3.2 Fixed broadband stratifies by geography

Most capital regions outperform their own country (Figure 1): Bangkok reaches 298.8 Mbps against a national 276.9, Kuala Lumpur 212.3 against 194.8, and Manila NCR 159.6 against 132.0. Singapore and Myanmar are the exceptions: Singapore’s Central Region (361.0) sits *below* its national mean (394.8), as does Yangon (22.0 against 24.6). The spread between the strongest and weakest capital region is over 16× (Singapore’s 361.0 against Yangon’s 22.0 Mbps), a gap that a table of national averages conceals.

The trajectory is also diverging rather than converging. Singapore nearly doubled its median provincial fixed speed over the period, from roughly 280 to 550 Mbps, and Thailand grew from roughly 200 to 300. Vietnam accelerated to join Malaysia above 200 Mbps. Laos, Cambodia, Indonesia and Myanmar remained below 60 Mbps for all three years.

The fixed–mobile gap also varies widely. Fixed broadband exceeds mobile by 3.2× in Thailand and 2.1× in Singapore, but the two are near parity in Malaysia and Indonesia (1.1×), and Myanmar is the only country in the set where mobile is *faster* than fixed (0.8×) — a mobile-first market.

3.3 UHD video separates the region

Raising the bar to 25 Mbps separates the region into two groups. Thailand, Vietnam, Singapore and Malaysia pass 100% of province-quarters, fixed and mobile. The Philippines is near-universal (99.6–100%), and Laos and Cambodia pass 88–99%. Indonesia and Myanmar diverge by connection type: Indonesia passes 88.1% fixed / 90.5% mobile, while Myanmar passes only 22.1% fixed but 68.8% mobile — the one country in the set where mobile UHD access is substantially *better* than fixed, consistent with its mobile-first fixed–mobile gap noted above.

Table 1: Test-weighted % of province-quarters clearing each cloud-gaming component (bandwidth ≥ 44 Mbps; Ookla latency ≤ 25 ms; both), 12 quarters, Q1 2023–Q4 2025.

Country	BW	Fixed			Mobile		
		BW	Lat	Both	BW	Lat	Both
Singapore	100.0	100.0	100.0	100.0	48.8	48.8	48.8
Vietnam	100.0	100.0	100.0	95.5	27.9	27.9	27.9
Malaysia	100.0	97.6	97.6	100.0	33.5	33.5	33.5
Thailand	100.0	99.5	99.5	99.4	21.6	21.6	21.6
Philippines	99.8	92.6	92.6	93.2	43.2	43.2	43.2
Cambodia	67.3	97.6	64.9	18.1	20.5	10.5	10.5
Laos	66.2	97.2	64.2	16.3	2.0	1.4	1.4
Indonesia	39.2	92.6	35.6	20.3	23.0	13.9	13.9
Myanmar	5.8	71.1	0.0	12.2	0.0	0.0	0.0

3.4 Cloud gaming: latency, not bandwidth, is the real constraint

Table 1 shows a binding constraint that flips by market tier and connection type. In the three lowest-tier fixed markets — Cambodia, Laos, Myanmar — bandwidth is what fails: latency clears 71–98% of the time, but bandwidth clears only 6–67%, capping the combined pass rate at 0–65%. On mobile, the pattern reverses *everywhere*: bandwidth clears 93–100% of province-quarters in every country except Cambodia, Laos and Myanmar, yet latency is what fails — including in Singapore, where mobile bandwidth clears 100% of the time but latency only 48.8%, and in Thailand, Vietnam and Malaysia, where mobile latency clears just 21.6–33.5%. A ranking built on download speed alone would call Singapore’s and Malaysia’s mobile networks cloud-gaming ready; once latency is included, they fail roughly half the time or worse.

4 ONGOING AND FUTURE WORK

Three directions are in progress. First, we are fitting province-level models of speed against economic and density covariates to separate infrastructure effects from wealth effects. Second, we are extending the NDT7 analysis to peak-hour behavior, where busy-versus-quiet *ratios* remain interpretable even though absolute latency is not, given offshore server placement. Third, we are quantifying the measurement bias introduced by thin-sample provinces, which currently inflate the apparent performance of several emerging markets.

The broader argument we wish to discuss at the poster session is that a speed index answers the wrong question, and that bandwidth alone answers only half of it. Basic voice and video access across Southeast Asia is effectively universal, and what separates fixed-broadband markets at the cloud-gaming tier is still mostly bandwidth. But for mobile networks in every country in the set, the constraint that actually blocks a latency-sensitive application is latency, not throughput — a distinction a download-speed ranking cannot see, and one that points at different infrastructure investment than a ranking does.

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